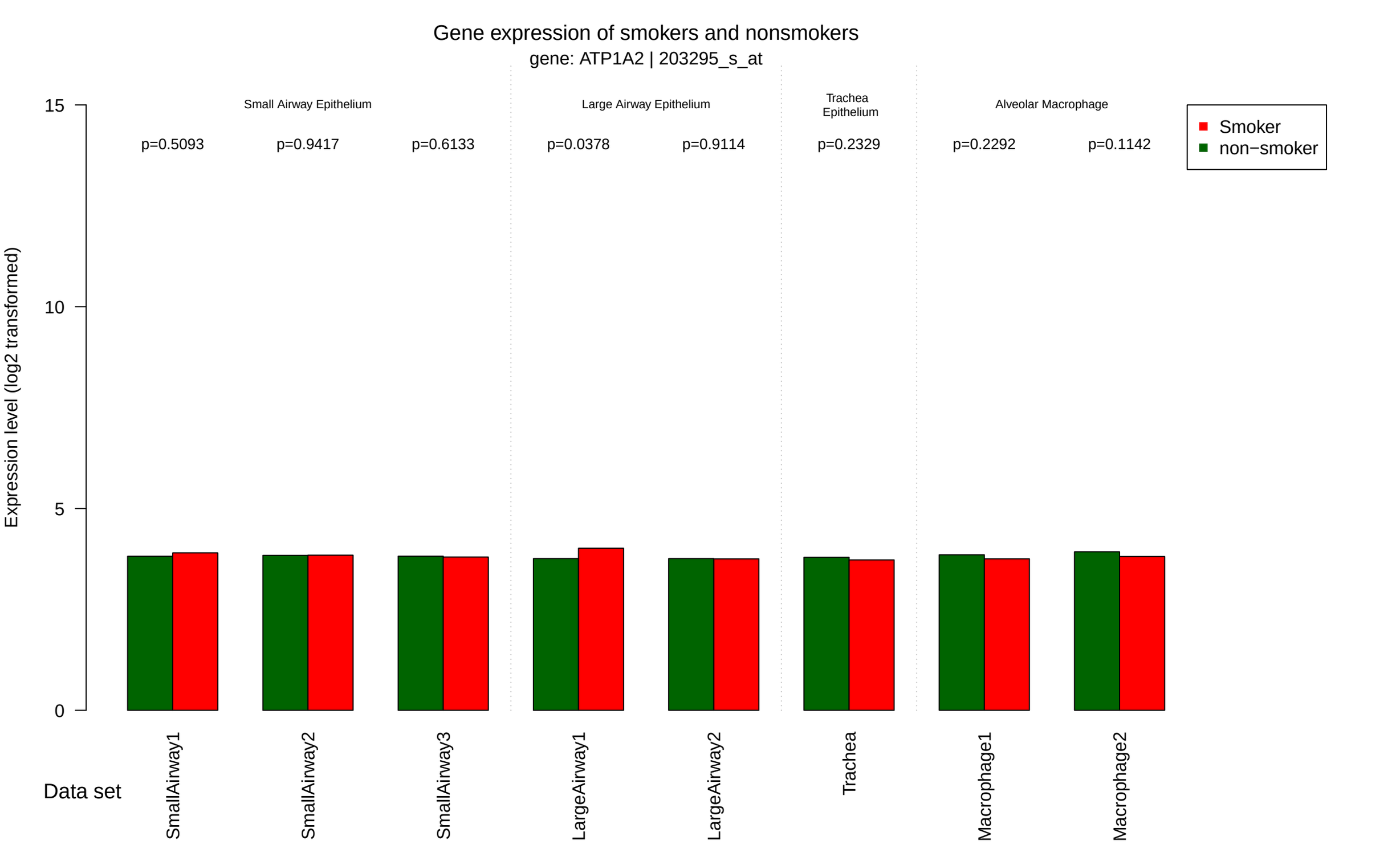
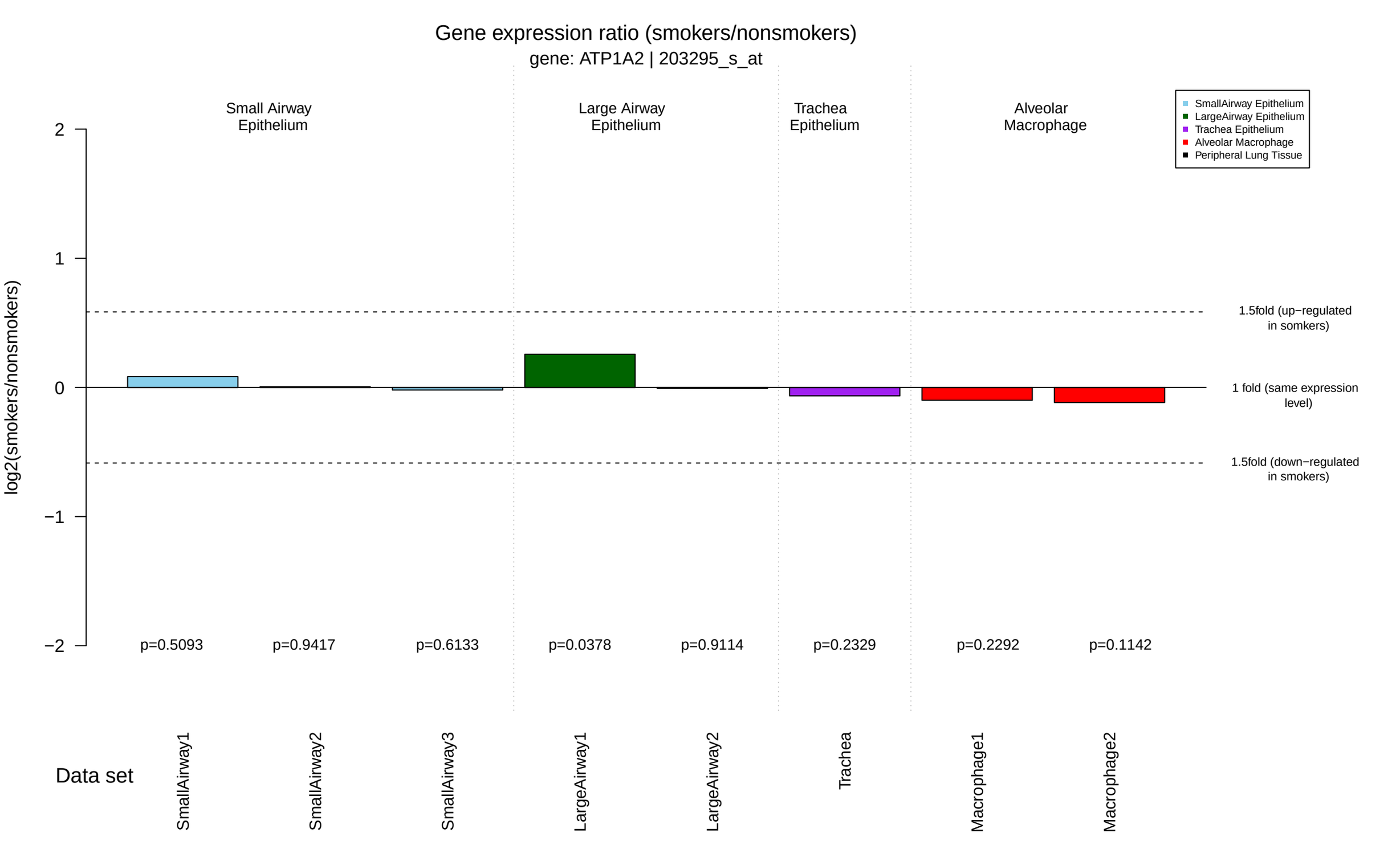
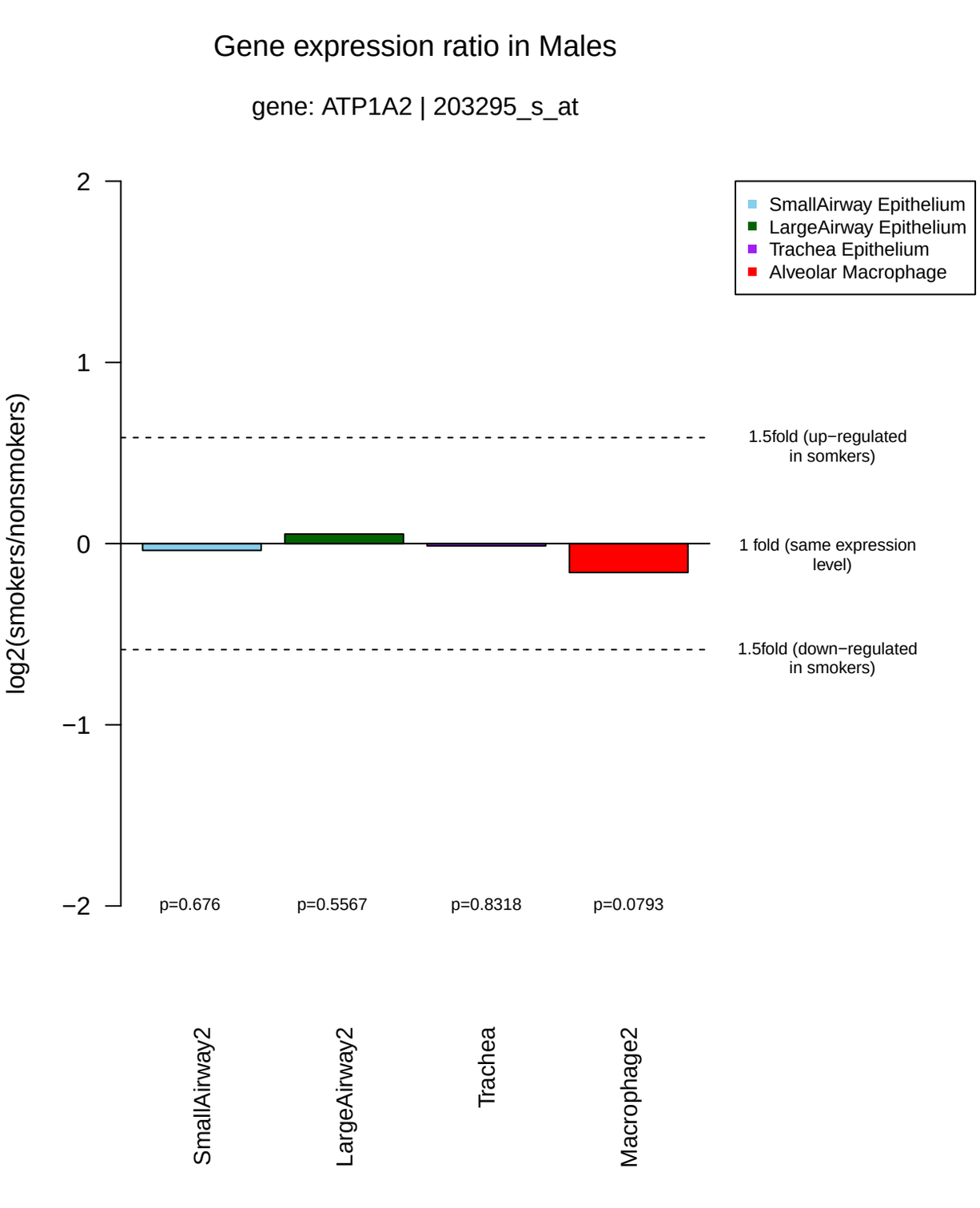
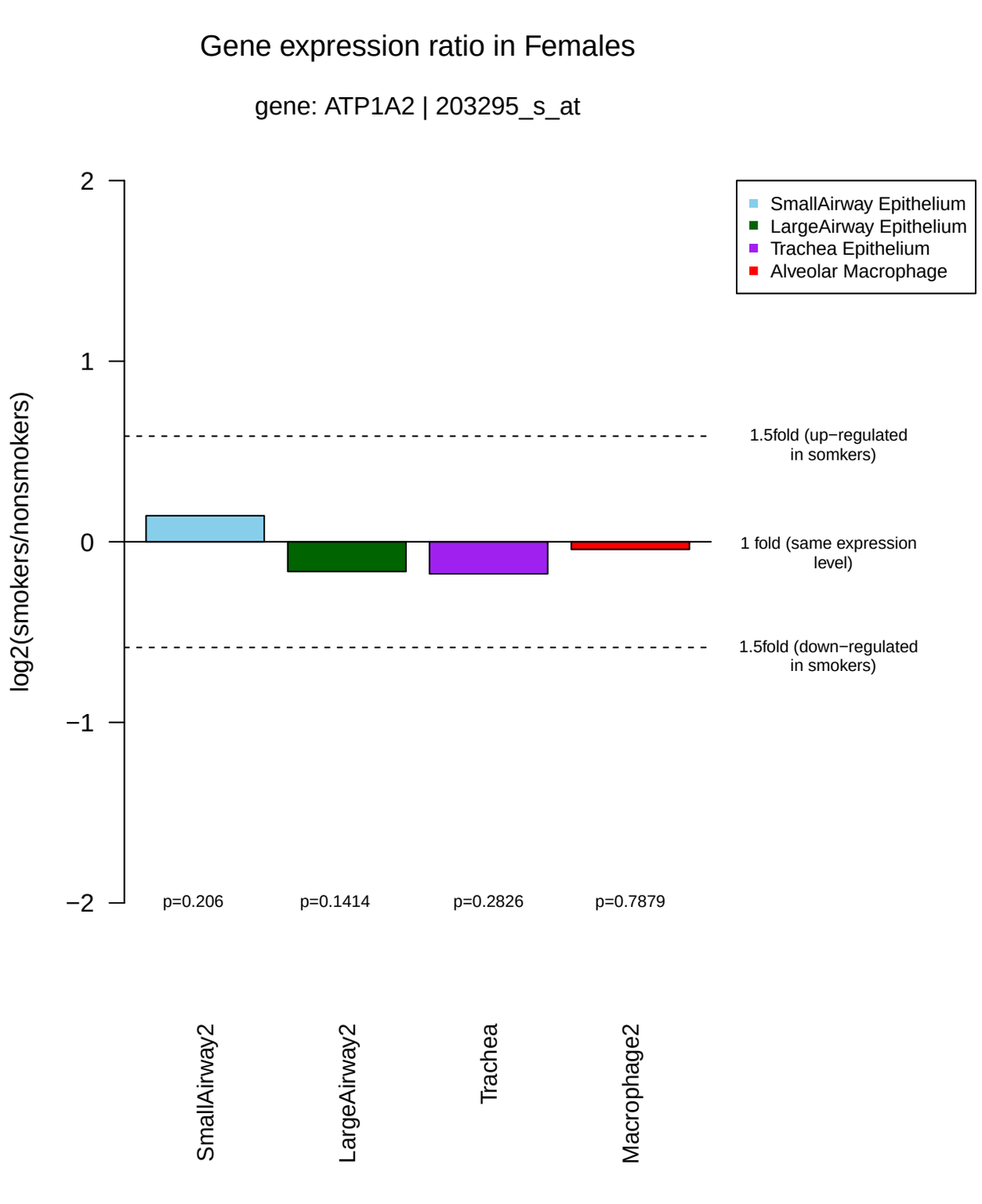
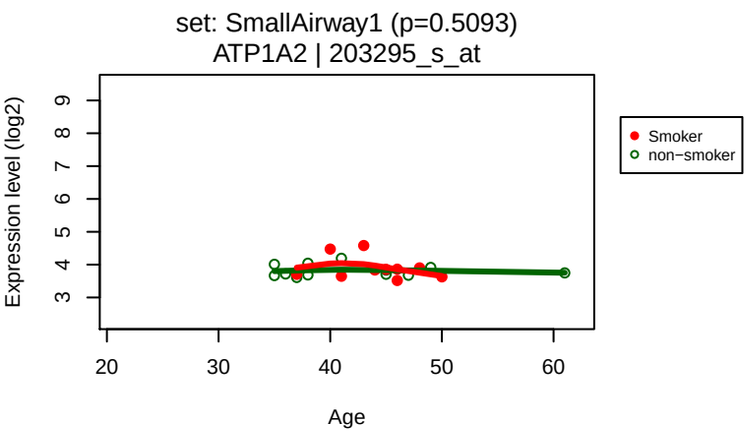
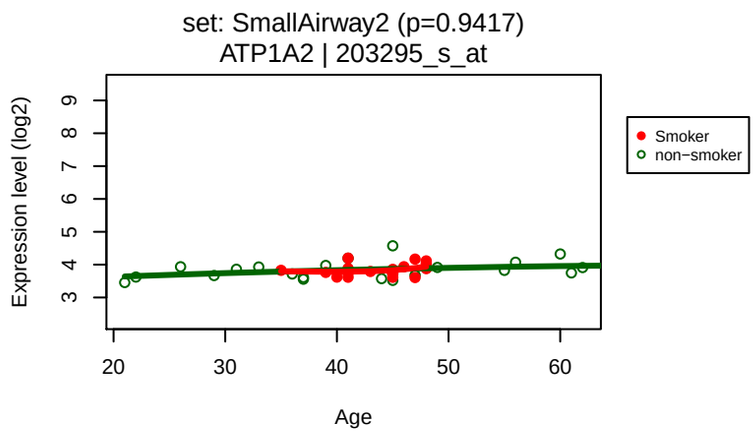

Gene expression of smokers and nonsmokers
gene: ATP1A2 | 203295_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5093
p=0.9417
p=0.6133
p=0.0378
p=0.9114
p=0.2329
p=0.2292
p=0.1142
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

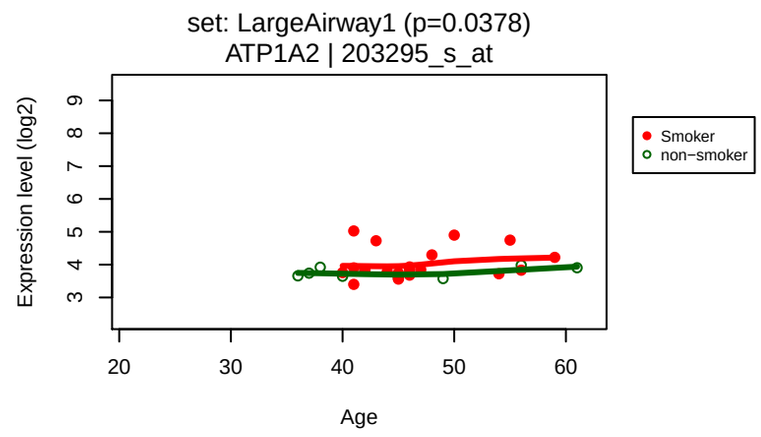

Gene expression ratio (smokers/nonsmokers)
gene: ATP1A2 | 203295_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5093
p=0.9417
p=0.6133
p=0.0378
p=0.9114
p=0.2329
p=0.2292
p=0.1142
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

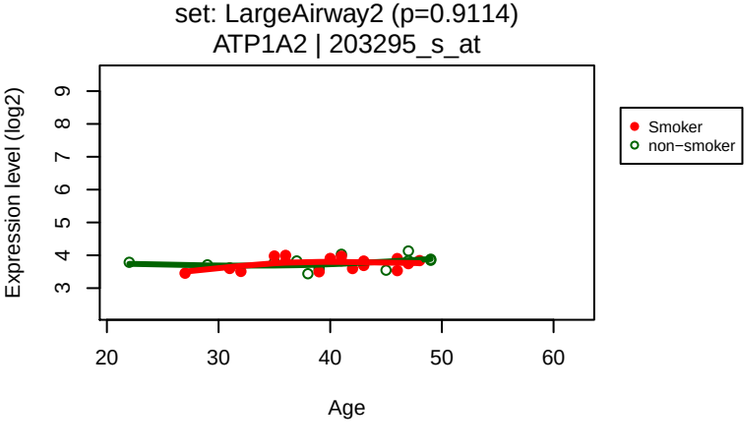

Gene expression ratio in Males
gene: ATP1A2 | 203295_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.676
p=0.5567
p=0.8318
p=0.0793
SmallAirway2
LargeAirway2
Trachea
Macrophage2

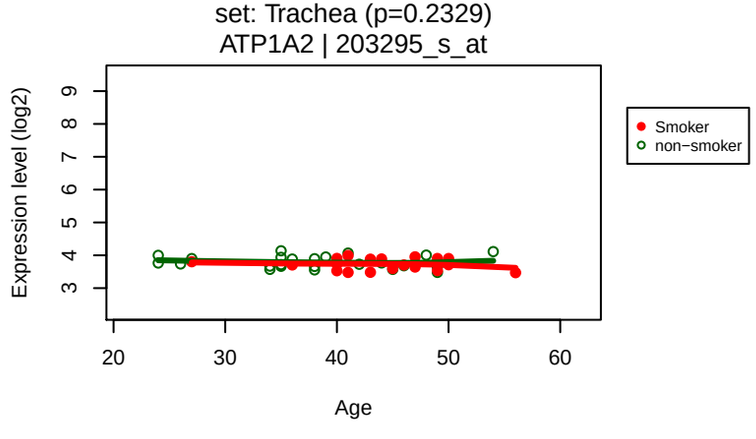

Gene expression ratio in Females
gene: ATP1A2 | 203295_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.206
p=0.1414
p=0.2826
p=0.7879
SmallAirway2
LargeAirway2
Trachea
Macrophage2

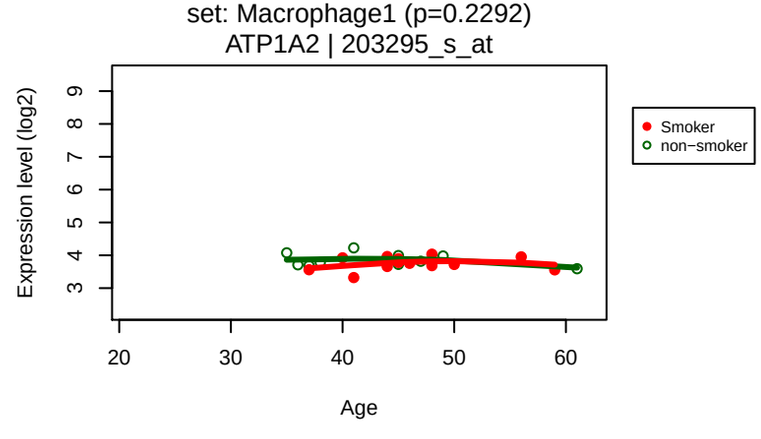

set: SmallAirway1 (p=0.5093)
ATP1A2 | 203295_s_at
Expression level (log2)
Age
Smoker
non−smoker

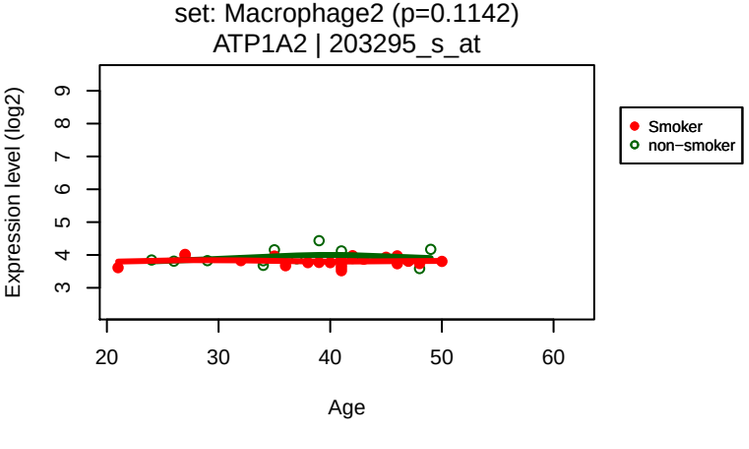

set: SmallAirway2 (p=0.9417)
ATP1A2 | 203295_s_at
Expression level (log2)
Age
Smoker
non−smoker

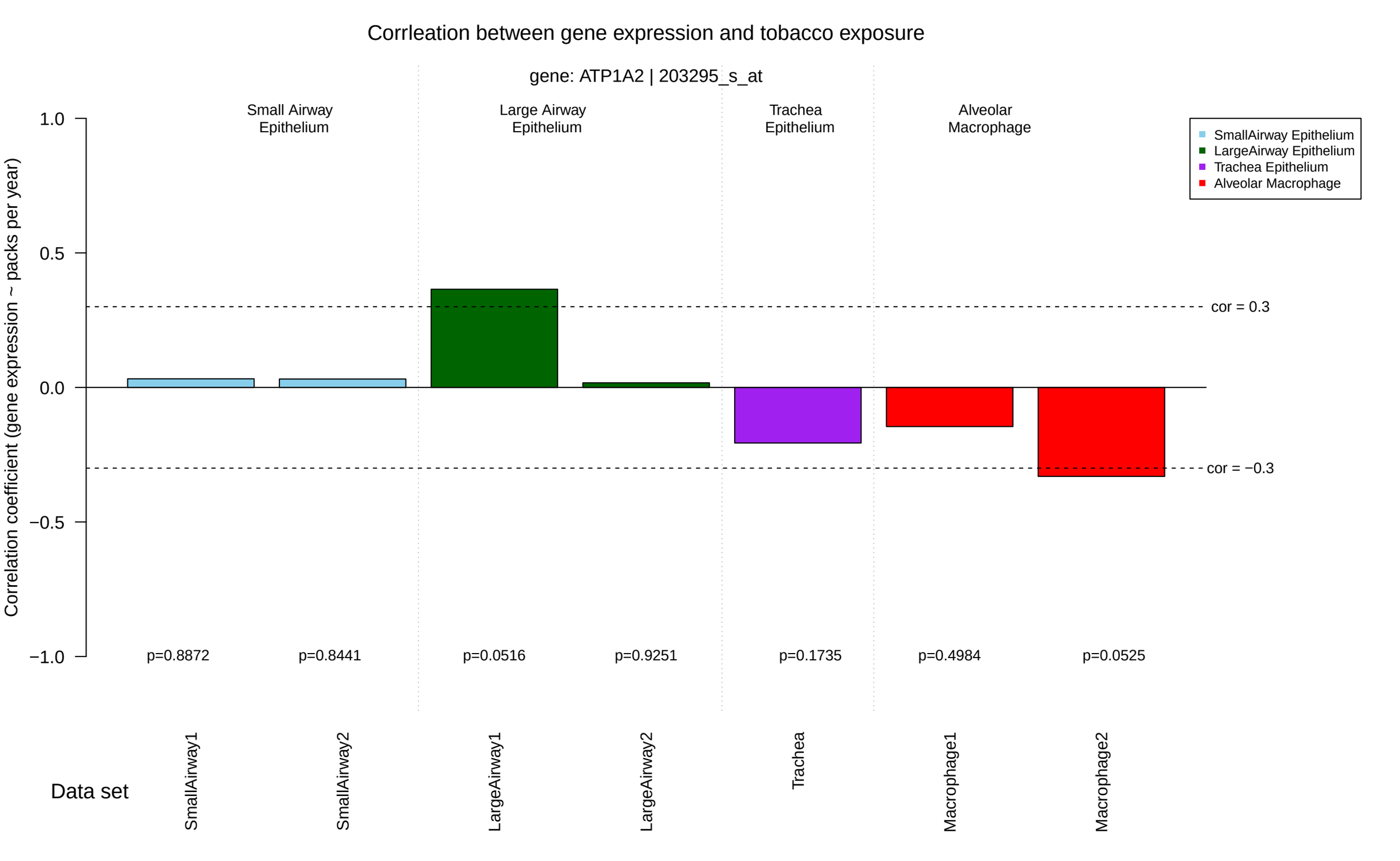

set: LargeAirway1 (p=0.0378)
ATP1A2 | 203295_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9114)
ATP1A2 | 203295_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2329)
ATP1A2 | 203295_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2292)
ATP1A2 | 203295_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1142)
ATP1A2 | 203295_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP1A2 | 203295_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.8872
p=0.8441
p=0.0516
p=0.9251
p=0.1735
p=0.4984
p=0.0525
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2